a

From Scratch | Foundation / zero-shot | FT-600K | FT-MultiT

Source trajectory stage

| | Lmax2 123 | Lmax2 234 | baseline 345 | OMAT legacy | MPA-0 medium | MP-0b3 medium | FT-600K 123 | FT-600K 234 | FT-600K 345 | FT-MultiT 123 | FT-MultiT 234 | FT-MultiT 345 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| early stable | 2.6 | 5.3 | 2.6 | 2.4 | 3.1 | 2.9 | 2.7 | 2.7 | 2.7 | 2.7 | 2.7 | 2.7 |
| pre-onset | 2.6 | 5.6 | 2.6 | 2.5 | 3.2 | 3.0 | 2.8 | 2.8 | 2.8 | 2.8 | 2.8 | 2.8 |
| precursor | 3.7 | 12.1 | 3.2 | 17.0 | 29.9 | 30.8 | 15.3 | 18.2 | 16.6 | 16.3 | 14.8 | 17.2 |
| threshold | 133.1 | 171.6 | 359.3 | 452.2 | 890.2 | 1298.8 | 893.2 | 488.6 | 918.3 | 1617.3 | 3295.0 | 2356.2 |
| stable terminal | 2.6 | 12.9 | 2.7 | 2.4 | 3.0 | 2.7 | 2.8 | 2.8 | 2.7 | 2.8 | 2.8 | 2.8 |

median max |F| (eV Å$^{-1}$)

b

static max |F| (eV Å$^{-1}$)

From Scratch
Foundation / zero-shot
FT-600K
FT-MultiT
arm 1
arm 2
arm 3
100 eV Å$^{-1}$ gate

early stable | pre-onset | precursor | threshold | stable terminal

source stage

MLIP checkpoint

c

SOAP PCA 2

neb
test
train
valid
precursor
early stable
pre-onset
stable terminal
threshold

SOAP PCA 1

d

12-model range of max |F| (eV Å$^{-1}$)

precursor
early stable
pre-onset
stable terminal
threshold

distance to DFT training distribution